A140 – Gravitation: G as a Bridge, and the QG Problem

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Contents

.1 The Claim

[STIPULATION] The gravitational constant G is in this theory not an independent constant of nature but a unit bridge – like c, $\hbar$, and k_B (A080).

From this follows the stronger and better-known claim of the corpus: the problem of quantum gravity does not arise in this form. This document examines what supports this claim and what does not.

.2 Why G Is a Bridge

[B] G connects a mass with a length. In the natural system of the theory, in which $\tilde{T} \cdot m = 1$ holds and time and mass are the same quantity in two readings, this connection does not need to be separately established – it is already in the duality.

This is the same statement as for $\hbar$: there the bridge between mass and time is meant, here the one between mass and length. Both are, in a system that does not carry time, mass, and length as independent units, conversions and not natural laws.

[STIPULATION] Operationally G remains indispensable. One who computes in SI needs it. Ontologically it is secondary. This is exactly the dual role that A080 describes for all bridge constants, and it is not a demotion.

.3 Why the QG Problem Does Not Arise in This Form

[B] The usual problem runs: quantise gravitation as a field and thereby obtain a renormalisable theory. It arises because the geometry is treated as a dynamic field that must in turn be quantised.

In this theory the geometry is *pre-ordered* and not dynamic in the same sense: the T^4/Z_3 is stipulated (A020), the quantum structure arises as its spectrum (A030, A070). There is therefore nothing left that would need to be subsequently quantised.

Honest assessment of this statement. It is not a solved problem but a *dissolved* one – and dissolved by a stipulation. One who does not stipulate the pre-ordered geometry still has the problem. The theory does not answer the question; it does not pose it.

This is the *negative* argument. A positive argument stands alongside it, which derives the numerical value of G from ξ: $G = \xi^2/(4\,m_{\text{char}})$ (A145). Both belong together – the negative explains why quantisation is unnecessary, the positive explains where the value comes from.

This is a legitimate move, but it must not be presented as a solution. The difference is the same as between a derivation and a calibration (A130): both are permissible, but they are not the same.

.4 What Follows from This and What Does Not

[B] It follows: no gravitons as necessary constituents, no renormalisation problems of gravitation, no singularities from a field quantisation.

[STIPULATION] It does *not* follow: that the classical gravitational dynamics is reproduced. The A-series shows nowhere that the observed gravitational interaction with its strength and inverse-square law follows from the compact geometry. This is a major open point and is not minimised here.

.5 Verification Script

- The bridge role of G: that G can be absorbed in natural units just like c, $\hbar$, and k_B, shown by the consistency of the unit equations: `python/A_Serie_Skripte/a140_g_bruecke.py`

.6 Open Questions

First, gravitational dynamics has not yet been treated as an A-series document – it is taken up in A142. The complete field-equation framework (time-field Lagrange, scalar/fermion coupling, modified Dirac equation, Higgs coupling, ξ from Higgs matching) is available in the legacy corpus and is incorporated in A142. What remains after that: (a) the complete quantisation of these field equations, (b) the transition to measurable predictions beyond the Schwarzschild case, (c) the connection to the projection chain (A090).

Second, the strength of the interaction is not derived. Treating G as a bridge does not explain why gravitation is forty orders of magnitude weaker than electromagnetism.

Third, the dissolution of the QG problem is a consequence of the stipulation. See above; booked as such, not as a result.

.7 Supersedes

This document subsumes: Dok. 012 (gravitational constant), Dok. 127 (gravitation), Dok. 163 (quantum gravity). Incorporated: P40.

.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.